ModelIt!

Teacher's Guide

Why Do Some Things Dissolve and Others Don't?

How Water Temperature and Motion Affect Dissolving Speed

Quick Reference

Grade Level	Grade 05L2
NGSS Standard	5-PS1-1, 5-PS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-PS1-1, 5-PS1-4

Develop a model to describe that matter is made of particles too small to be seen. Conduct an investigation to determine whether the mixing of two or more substances results in new substances.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how temperature and stirring affect particle interactions and dissolving.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter** — Students model how particle-level interactions explain observable dissolving behavior and solution formation.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between temperature, stirring, particle contact, and dissolving speed.

CAST Testing Alignment

MULTIPLE CHOICE:

A student puts identical sugar cubes in two cups of water. Cup A is hot and stirred. Cup B is cold and not stirred. Which sugar cube dissolves first and what is the MAIN reason?

CONSTRUCTED RESPONSE:

Explain why hot, stirred water dissolves sugar faster than cold, still water. Use the concept of particle contact rate from your model in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Dissolving is a physical process where a solid breaks apart into particles so small they spread evenly throughout a liquid, forming a solution. Understanding what controls dissolving speed requires thinking about what happens at the particle level — even though we cannot see individual particles.

What Happens When Something Dissolves?

When a solid dissolves, its particles separate from each other and mix evenly among the liquid particles. The solid does not disappear — its particles are still there, just spread out. You can prove this by evaporating the water and finding the solid left behind.

Temperature and Particle Motion

In warmer water, particles move faster. These fast-moving water particles collide with the solid surface more often and with more energy, breaking apart the solid particles more quickly. This is why hot water dissolves most substances faster than cold water.

Stirring and Fresh Contact

When water is still, dissolved particles accumulate near the solid surface, creating a 'barrier' of already-saturated water. Stirring moves this saturated water away and brings fresh water into contact with the solid, maintaining a high rate of dissolving.

Saturation: The Dissolving Limit

Every liquid can only dissolve a certain amount of a substance at a given temperature. When no more can dissolve, the solution is saturated. Warming the water usually increases how much can dissolve, which is why more sugar dissolves in hot tea than iced tea.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Water Temperature, Stirring Speed, Particle Contact Rate, Dissolving Speed, and Solution Concentration as the five system components.

E **Establish Relationships**

Students determine that Water Temperature increases Particle Contact Rate (+), Stirring Speed increases Particle Contact Rate (+), Particle Contact Rate increases Dissolving Speed (+), and Dissolving Speed increases Solution Concentration (+).

V **Visualize & Model**

Students build the dissolving model showing how external conditions (temperature and stirring) drive particle-level interactions that produce observable dissolving behavior.

E **Evaluate Outcomes**

Students run the Hot Water, No Stirring scenario and compare with the full hot-stirred scenario to isolate the effect of stirring alone, then test cold-no-stirring to see both factors removed.

R **Revise & Extend**

Students add Particle Size or Substance Type to explore how physical properties of the solid affect dissolving.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You have done this a hundred times — stirred sugar into a drink. But do you know WHY it dissolves?"*

Watch For

Drop a sugar cube into a glass of water in front of the class. Ask: What is happening?

Slide 2: Learning Objectives

2 min

Say: *"Today we model the invisible particle collisions that make dissolving happen."*

Watch For

Students read objectives and discuss: Where does the sugar GO?

Slide 3: Big Question

2 min

Say: *"Sugar dissolves. Sand does not. Hot water works faster than cold. Let's figure out why."*

Watch For

Quick-write: List three things you could do to make sugar dissolve faster.

Slide 4: LEVER Framework

2 min

Say: *"Our model has two external factors that BOTH affect the same internal process — particle contact rate."*

Watch For

Preview: temperature and stirring both increase particle collisions.

Slide 5: Activity 1: Components

8 min

Say: *"Water Temperature and Stirring Speed are external — we control them. The other three respond."*

Watch For

Students sort component cards into external and internal categories.

Slide 6: Activity 2: Connections

8 min

Say: *"Notice that BOTH external components feed into Particle Contact Rate — it is the bottleneck of the system."*

Watch For

Guide students to draw all four relationship arrows in ModelIt.

Slide 7: Activity 3: Simulation

10 min

Say: *"Hot and stirred vs cold and still — how big is the difference? Let's race them."*

Watch For

Students run all three scenarios and record observations.

Slide 8: Discoveries

5 min

Say: *"Everything comes down to particle contact. More collisions, faster dissolving."*

Watch For

Class discussion: Where do you see dissolving in everyday life?

Slide 9: STEM Challenge

10 min

Say: *"Design the fastest method to dissolve a sugar cube. Time it. Beat the other teams."*

Watch For

Groups design methods, test with real materials, and record times.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Water Temperature, Stirring Speed
- **INTERNAL:** Particle Contact Rate, Dissolving Speed, Solution Concentration

Relationships

- **Water Temperature to Particle Contact Rate:** POSITIVE (+) — Warmer water particles move faster, colliding with the solid surface more frequently.
- **Stirring Speed to Particle Contact Rate:** POSITIVE (+) — Stirring brings fresh water particles into contact with the solid, increasing collision frequency.
- **Particle Contact Rate to Dissolving Speed:** POSITIVE (+) — More frequent particle collisions break apart the solid faster.
- **Dissolving Speed to Solution Concentration:** POSITIVE (+) — Faster dissolving increases the amount of dissolved substance in the water over time.

Key Discoveries

- Hot water dissolves substances faster because its particles move faster and collide with the solid more often
- Stirring increases dissolving speed by bringing fresh water particles into contact with the solid surface
- BOTH temperature and stirring increase particle contact rate, which is the key to dissolving speed
- Solution concentration builds over time as more and more of the solid dissolves into the water

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Dissolved substances disappear"	When a solid dissolves, it does not vanish. Its particles separate and mix evenly with the water particles. The dissolved substance is still there — the water tastes sweet, weighs more, and the solid reappears if you evaporate the water.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Stirring creates heat that dissolves things"	Stirring does not significantly heat the water. It works by moving fresh, unsaturated water to the solid surface. The mechanical motion replaces the saturated layer near the solid with water that can still dissolve more material.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Everything dissolves if you stir long enough"	Some substances simply do not dissolve in water regardless of temperature or stirring. Sand, plastic, and oil do not dissolve because their particles are not attracted to water particles. Solubility is a property of the substance itself.